10.1.1 Types of services

I will be using the terms *service* and *API* interchangeably in this chapter. What I really mean in either case is the API of a synchronous service, asynchronous services, or *sui generis* (unique) service types. Let's have a closer look at the types of services now:

- *Synchronous services*, generally request–response-based systems, such as long-running RESTful web services, WebSocket-based systems, and RPC-based (gRPC or SOAP) systems but also data pipelines (e.g., ETL)
- *Asynchronous services*, including event buses, queues, and Pub/Sub systems in which there is no obvious causal relation between a publishing action, such as *put event X on a queue with topic Z*, and a consuming action, like *read the most recent event from queue with topic Z*
- *Sui generis services*, including
  - Batch jobs triggered by a certain event, run for a certain time, and expected to terminate
  - Storage, such as POSIX-compliant local or remote file systems (including `ext4` and NFS) or object storage, like Amazon S3
  - Databases and datastores, including relational databases, such as PostgreSQL, and NoSQL datastores, like AWS DynamoDB or MongoDB

Depending on the type of service, there are different ways to measure how the service customer perceives the service. Enter SLIs.

10.1.2 Service level indicator

The service level indicator (SLI) is a quantitative means to determine how your service is doing. In short, all SLIs are metrics, but not all metrics are SLIs. The purpose of an SLI is to allow you to measure how the service customer is experiencing your service, based on a numerical value. You can aggregate this value and do fancy math, like percentiles, on it. But at the end of the day, SLIs really are only service metrics, such as

- Uptime, availability, or reachability of a service
- Error rate (e.g., errors to total requests) of a synchronous service
- Duration, or latency, such as how long a synchronous service took to respond to a request
- Successfully completed or failed jobs for an asynchronous service
- Throughput, or how many transactions per second (TPS) a service is able to sustain

If you're interested in details on what kind of metrics (for synchronous services) to consider, I recommend reading "The RED Method: A New Approach to Monitoring Microservices" by Joab Jackson (http://mng.bz/a1nj). Now that you know what an SLI is on a high level, a service metric that you, as an engineer, define and track, let's move on to setting goals—or SLOs.